



Practice assignment 8 - Not Graded



This assignment will not be graded and is only for practice.

Note: Suppose W_1 and W_2 are subspaces of a vector space V . Then sum of two subspaces is defined as:

$$W_1 + W_2 = \{w_1 + w_2 \mid w_1 \in W_1, w_2 \in W_2\}.$$

One can verify that $W_1 + W_2$ is a subspace of V .

Multiple Choice Questions (MCQ):

1 point

An inner product on R^3 is defined as:

$$\langle \cdot, \cdot \rangle: R^3 \times R^3 \rightarrow R$$

$$\langle (x_1, x_2, x_3), (y_1, y_2, y_3) \rangle = x_1 y_1 + x_2 y_2 + x_3 y_3.$$

Match a set S in Column A with the set T given in column B, so that $T \subset \text{Span}(S)^\perp$. Match a set in column B with the property it satisfies in column C.



Which of the following option are true?

☐

d → ii → 3

☐

d → iii → 4

☐

c → ii → 4

☐

a → iv → 1

☐

a → ii → 3

☐

b → iv → 2

No, the answer is incorrect.

Score: 0

Accepted Answers:

d → iii → 4

a → ii → 3

Suppose W_1 and W_2 W_1 and W_2 are subspaces of a vector space V . Let P_{W_1} and P_{W_2} P_{W_1} and P_{W_2} denote the projection from V to W_1 and V to W_2 respectively and consider the following statements:

- **Statement P:** If $P_{W_1} + P_{W_2}$ $P_{W_1} + P_{W_2}$ is a projection from V to $W_1 + W_2$, then $P_{W_1} \circ P_{W_2} + P_{W_2} \circ P_{W_1} = 0$. $P_{W_1} \circ P_{W_2} + P_{W_2} \circ P_{W_1} = 0$.
- **Statement Q:** $P_{W_1}^2 := P_{W_1} \circ P_{W_1}$ $P_{W_1}^2 := P_{W_1} \circ P_{W_1}$ is not a projection from V to W_1 .

- **Statement R:** If AA is the matrix representation of $P_{W_2} P_{W_1}$, then AA can not be a symmetric matrix.
- **Statement S:** $P_{W_1} - P_{W_2} P_{W_1} P_{W_2}$ is a projection from V to $W_1 + W_2$.

Find the number of correct statements

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 1

1 point

1 point

Which of the followings option(s) is (are) true?

☐

Let u, v , and w be three vectors in R^2 and $\langle \cdot, \cdot \rangle$ be an inner product on R^2 . If $\langle u, v \rangle = \langle u, w \rangle$ and $\langle v, w \rangle = \langle u, w \rangle$, then $v = w$

☐

Let V be an inner product space. If $\langle u, v \rangle = 0$ for all $u \in V$, then $v = 0$

☐

Let V be an inner product space and $u, v \in V$. If $\langle u + v, u - v \rangle = 0$, then $\|u\| \neq \|v\|$

☐

Let V be an inner product space. If $S \subset V$ then $S^\perp = (\text{span}(S))^\perp$

☐

Suppose $\{u_1, u_2, \dots, u_m\}$ be an orthogonal set of vectors in an inner product space V , then $\|u_1 + u_2 + \dots + u_m\|^2 = \|u_1\|^2 + \|u_2\|^2 + \dots + \|u_m\|^2$

No, the answer is incorrect.

Score: 0

Accepted Answers:

Let V be an inner product space. If $\langle u, v \rangle = 0$ for all $u \in V$, then $v = 0$

Let V be an inner product space. If $S \subset V$ then $S^\perp = (\text{span}(S))^\perp$

Suppose $\{u_1, u_2, \dots, u_m\}$ be an orthogonal set of vectors in an inner product space V , then $\|u_1 + u_2 + \dots + u_m\|^2 = \|u_1\|^2 + \|u_2\|^2 + \dots + \|u_m\|^2$

1 point

Which of the following option(s) is/are true?

☐

Let A and B be orthogonal matrices of order 3. Then A and BA and B are equivalent matrices.

☐

Let A and B be orthogonal matrices. Then A and BA and B are similar matrices.

☐

Let A be a rotation matrix corresponding to the anti-clock wise rotation of the XY-plane about the Z-axis with angle θ . Then nullity of the matrix A is 0.

☐

Let A be a rotation matrix corresponding to the anti-clock wise rotation of the XY-plane about the Z-axis with angle θ and B be a rotation matrix corresponding to the anti-clock wise rotation of the YZ-plane about the X-axis with angle β . Then A and B are equivalent matrices.

No, the answer is incorrect.

Score: 0

Accepted Answers:

Let A and B be orthogonal matrices of order 3. Then A and BA and B are equivalent matrices.

Let A be a rotation matrix corresponding to the anti-clock wise rotation of the XY -plane about the Z -axis with angle θ . Then nullity of the matrix A is 0.

Let A be a rotation matrix corresponding to the anti-clock wise rotation of the XY -plane about the Z -axis with angle θ and B be a rotation matrix corresponding to the anti- clock wise rotation of the YZ -plane about the X -axis with angle β . Then A and B are equivalent matrices.

Numerical Answer Type:

Let $u = (1, 2, 1)$ and $v = (1, 2, 3)$ be the vectors of the inner product space R^3 with usual inner product. Let $T: R^3 \rightarrow R^3$ be an orthogonal linear transformation. If θ is an angle between $T(u)$ and $T(v)$, then find the value of $||T(u)|| ||T(v)|| \cos \theta$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 8

1 point

Consider a set $W = \{(1, 1, 1)\}$ from the inner product space R^3 . Find the dimension of the subspace $W^\perp$, where $W^\perp$ is the collection of the vectors which are orthogonal to the vector $(1, 1, 1)$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 2

1 point

Let $v = (3, 1, 2)$ be a vector in R^3 . If (a, b, c) is the vector obtained from v after the anti- clock wise rotation of ZX -plane with angle 45° about the Y - axis, then find the value of $\sqrt{2}(a + b + c - 1)$.

$(a + b + c - 1)$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 6

1 point

Comprehension Type Question:

With a particular frame of reference (in R^2) the position of Rahul's house is at the point $(8, 1)$. There are two roads Road 1 and Road 2 along the lines $y = 2x$ and $y = -2x$ respectively on the XY -plane.

Answer questions 8,9 and 10 using the given information.

1 point

Which of the following option(s) is (are) true?

☐ If Rahul wants to travel the minimum distance from his house to Road 1, then he will meet Road 1 at (2,4).

☐

If Rahul wants to travel the minimum distance from his house to Road 1, then he will meet Road 1 at $\frac{6}{5}(1, -2)$ $\frac{6}{5}(1, -2)$.

☐

If Rahul wants to travel the minimum distance from his house to Road 2, then he will meet Road 2 at $\frac{6}{5}(1, -2)$ $\frac{6}{5}(1, -2)$.

☐ If Rahul wants to travel the minimum distance from his house to Road 2, then he will meet Road 2 at (2,4).

No, the answer is incorrect.

Score: 0

Accepted Answers:

If Rahul wants to travel the minimum distance from his house to Road 1, then he will meet Road 1 at (2,4).

If Rahul wants to travel the minimum distance from his house to Road 2, then he will meet Road 2 at $\frac{6}{5}(1, -2)$ $\frac{6}{5}(1, -2)$.

Let θ be the acute angle between Road 1 and Road 2, then what will be the value of $5\cos(\theta)$ $5\cos(\theta)$?

No, the answer is incorrect.

Score: 0